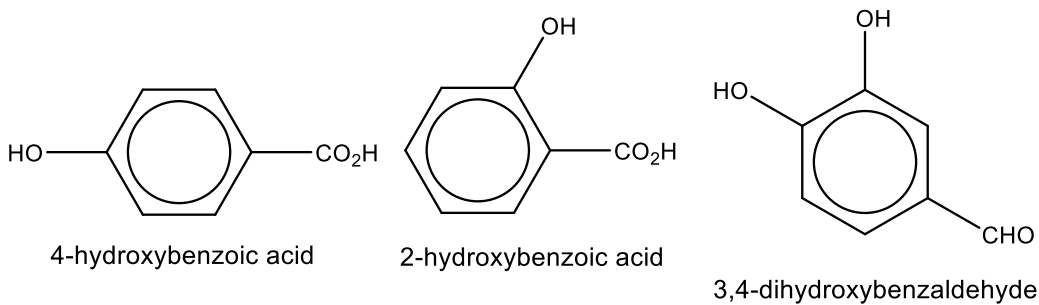


- 10 4-hydroxybenzoic acid (A), 2-hydroxybenzoic acid (B) and 3,4-dihydroxybenzaldehyde (C) share the same molecular formula.



All three compounds combust exothermically. Their standard enthalpy changes of formation are tabulated below.

	standard enthalpy change of formation / kJ mol^{-1}
4-hydroxybenzoic acid	-481
2-hydroxybenzoic acid	-493
3,4-dihydroxybenzaldehyde	-392

Which statements are correct?

- 1 2-hydroxybenzoic acid and 3,4-dihydroxybenzaldehyde are chain isomers.
- 2 The magnitude of the standard enthalpy change of combustion decreases in the order $C > A > B$.
- 3 The thermodynamic stability decreases in the order $B > A > C$.

A 1, 2 and 3 **B** 1 and 2 **C** 1 and 3 **D** 2 and 3

- 11 A 0.483 g sample of glycine ($M_r = 75.0$) was placed in a bomb calorimeter and then ignited in